



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

SEPTEMBER 2007

Yearly Examination

YEAR 11

Mathematics Extension (Continuers)

General Instructions

- Reading Time – 5 Minutes.
- Working time – 60 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Start each NEW section in a separate answer booklet.

Total Marks – 60

- Attempt questions 1 – 4

Examiner: R. Boros

Total marks 60

Attempt questions 1 to 4

Answer each **Section** in a **Separate** writing booklet

Section A (Use a SEPARATE writing booklet)

Question 1 (14 marks)

- (a) Find the acute angle between the lines $y = 2x + 1$ and $y = -x + 1$, correct to the nearest minute. 2
- (b) Consider the polynomial $K(x) = 4x^3 + tx^2 + 2x - 1$. Given that $x + 1$ is a factor of $K(x)$, find the value of t . 2
- (c) The parametric equations of a curve are $x = \frac{2}{t}$ and $y = 2t^2$. What is the cartesian equation for the curve? 2
- (d) For the parabola $(x - 3)^2 = 6y + 12$, find the:
- i. coordinates of the vertex 1
 - ii. coordinates of the focus 1
 - iii. equation of the directrix 1
- (e) Find the coordinates of the point Q which divides the interval joining $A(2, -3)$ and $B(-4, 1)$ externally in the ratio $1 : 3$. 3
- (f) Sketch the graph of $y = x^2(x - 2)^3$ without the use of calculus. 2

Question 2 (14 marks)

(a) Differentiate $f(x) = 5 - x^2$ by using first principles. 2

(b) Given that $\sin \alpha = \frac{7}{25}$ and $\cos \beta = -\frac{3}{5}$ where α and β are obtuse angles, find the exact value of:

i. $\sin 2\alpha$ 1

ii. $\cos(\alpha + \beta)$ 2

(c) In a class of 30 students, 22 study Chemistry, 18 study Physics and 13 study both Chemistry and Physics. If a student is chosen at random, what is the probability that the student studies Chemistry or Physics? 2

(d) i. Express $\sin x - \sqrt{3} \cos x$ in the form $A \sin(x - \alpha)$, with $A > 0$ and $0 < \alpha < \frac{\pi}{2}$. 2

ii. Find the solutions to $\sin x - \sqrt{3} \cos x = \frac{2}{\sqrt{2}}$ for $0 \leq x \leq 2\pi$. 2

(e) Solve the inequation $\frac{1}{x+2} \leq \frac{1}{x+3}$. 3

Section B (Use a SEPARATE writing booklet)

Question 3 (15 marks)

(a) Solve $\log_{27} 16 = x \log_3 2$. 2

(b) Prove the trigonometric identity $\frac{\cos 2x}{(\cos x + \sin x)^3} = \frac{\cos x - \sin x}{1 + \sin 2x}$. 3

(c) A committee of three is to be chosen from a group of four males and five females. 2
The committee must include at least one male and at least one female. How many different committees can be formed?

(d) A maths teacher pays \$1000 into a superannuation fund at the beginning of each year. Compound interest is paid at 9%p.a. on the investment.

i. Show that the first \$1000 invested becomes \$20413.97 to the nearest cent after 35 years. 1

ii. What will be the value of the investment at the end of 35 years? Answer correct to the nearest dollar. 3

(e) Given that the cubic equation $2x^3 + 6x - 1 = 0$ has real roots α , β and γ . Evaluate:

i. $\alpha^3\beta^3\gamma^2 + \alpha^3\beta^2\gamma^3 + \alpha^2\beta^3\gamma^3$. 2

ii. $\frac{\alpha}{\beta\gamma} + \frac{\beta}{\alpha\gamma} + \frac{\gamma}{\alpha\beta}$. 2

Question 4 (17 marks)

(a) The derivative of $x\sqrt{x^2 + 3}$ is $\frac{ax^2 + b}{\sqrt{x^2 + 3}}$, where a and b are constants. Find the value of a and b . 3

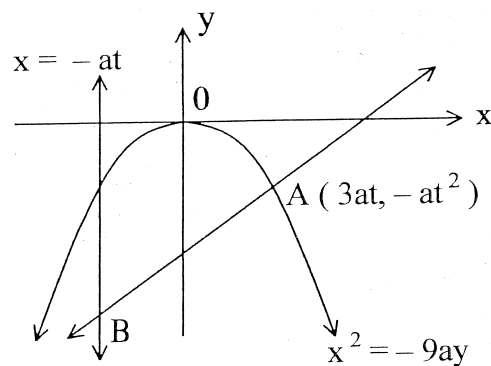
(b) The student council at a local school consists of 4 boys and 2 girls. In how many ways can they sit next to each other around a circular table for a meeting if:

i. there are no restrictions. 1

ii. the girls are not to sit next to each other. 2

(c) Find the general solution of the equation $\tan 2\theta = \tan \theta$ in radians. 3

(d)



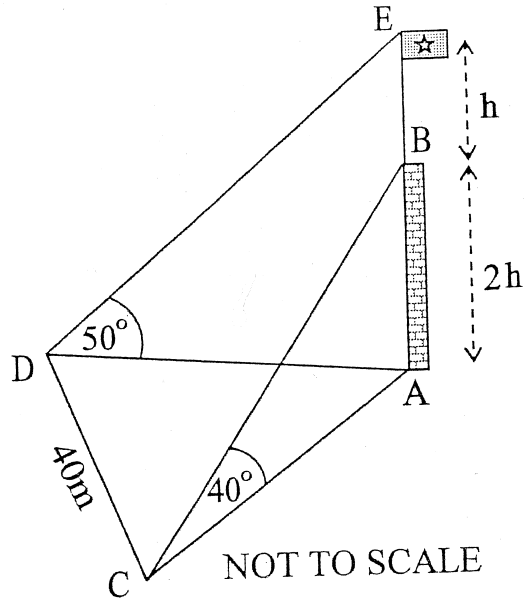
The point $A(3at, -at^2)$ is a variable point on the parabola $x^2 = -9ay$. The normal at A meets the line $x = -at$ at the point B .

i. Show that the equation of the normal to the parabola at A is 2

$$3x - 2ty = 2at^3 + 9at.$$

ii. Find the coordinates of B . 1

(e)



A building AB of height $2h$ metres has a flag pole of height h metres on top of it. From a point C , due south of the building, the angle of elevation of the top of the building is 40° . From a point D , due west of the building, the angle of elevation of the top of the flagpole is 50° . The points C and D are on the same level as A and they are 40 metres apart.

- Find expressions for AC and AD in terms of h .
- Show that $h = \frac{40}{\sqrt{4 \cot^2 40^\circ + 9 \cot^2 50^\circ}}$.
- Find to the nearest degree, the true bearing of D from C .

2

2

1

End of paper

Section A
Question 1.

$$(a) \tan \alpha = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$
$$= \left| \frac{2 - (-1)}{1 + (2)(-1)} \right|$$

$$= 3.$$

$$\tan \alpha = 3$$

$$\alpha = 71^\circ 34'$$

$$(b). K(-1) = -4 + t - 2 - 1; = 0.$$

$$t = 7.$$

$$(c) x = \frac{2}{t} \Rightarrow t = \frac{2}{x}.$$

So

$$y = 2 \left(\frac{2}{x} \right)^2$$

$$y = \frac{8}{x^2}.$$

$$(d) (x-3)^2 = 4 \left(\frac{3}{2} \right) (y+2).$$

(i) Vertex $(3, -2).$

(ii) Focus $(3, -\frac{1}{2}).$

(iii) Directrix $y = -\frac{7}{2}$ or $-3\frac{1}{2}$

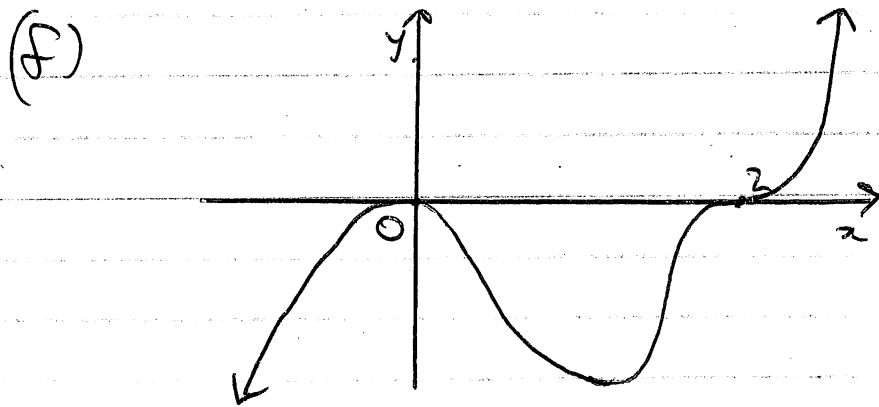
$$(e) \left(\frac{nx_1 + mx_2}{m+n}, \frac{ny_1 + my_2}{m+n} \right)$$

External division $m:-n$

$$\left(\frac{-3 \times 2 + 1 \times -4}{1-3}, \frac{-3 \times -3 + 1 \times 1}{1-3} \right)$$

$$= \left(\frac{6-4}{-2}, \frac{9+1}{-2} \right)$$

$$= (-1, -5)$$



QUESTION 2.

$$(a) f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{5 - (x+h)^2 - (5 - x^2)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{5 - x^2 - 2xh - h^2 - 5 + x^2}{h}$$

$$= \lim_{h \rightarrow 0} -2x + h$$

$$= -2x$$

$$(b) \sin \alpha = \frac{7}{25} \Rightarrow \cos \alpha = -\frac{24}{25}$$

$$\cos \beta = -\frac{3}{5} \Rightarrow \sin \beta = \frac{4}{5}$$

$$(i) \sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$= 2 \times \frac{7}{25} \times -\frac{24}{25}$$

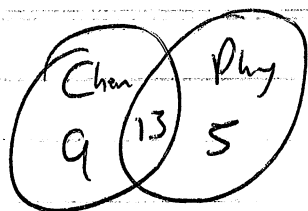
$$= -\frac{336}{625}$$

$$(ii) \cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$= -\frac{24}{25} \times -\frac{3}{5} - \frac{7}{25} \times \frac{4}{5}$$

$$= \frac{44}{125}$$

(c)



$$\frac{27}{30} = \frac{9}{10}.$$

(d)(i) $A \sin(x - \alpha) = A \cos \alpha \sin x - A \sin \alpha \cos x.$

So $A \cos \alpha = 1$
 $A \sin \alpha = \sqrt{3}.$

Thus $\tan \alpha = \sqrt{3}.$
 $\alpha = \frac{\pi}{3}$

and $A^2 \cos^2 \alpha + A^2 \sin^2 \alpha = 1 + 3.$
 $A^2 = 4$
 $A = 2.$

$$\sin x - \sqrt{3} \cos x \equiv 2 \sin \left(x - \frac{\pi}{3} \right).$$

(ii). $2 \sin \left(x - \frac{\pi}{3} \right) = \frac{2}{\sqrt{2}}$

$$\sin \left(x - \frac{\pi}{3} \right) = \frac{1}{\sqrt{2}}$$

$$x - \frac{\pi}{3} = \frac{\pi}{4}, \frac{3\pi}{4}$$

$$x = \frac{7\pi}{12}, \frac{13\pi}{12}.$$

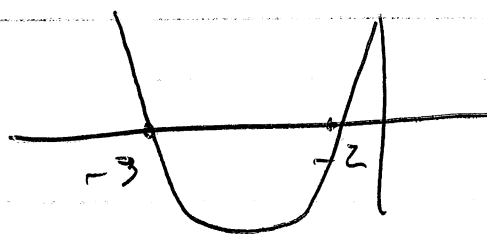
(e) $\frac{1}{x+2} - \frac{1}{x+3} \leq 0.$

$$\frac{(x+3) - (x+2)}{(x+3)(x+2)} \leq 0$$

$$\frac{1}{(x+3)(x+2)} \leq 0.$$

$$(x+3)(x+2) \leq 0.$$

$$-3 \leq x \leq -2.$$



B.1 2007 Yr 11 Yearly - Continuers SECTION B

QUESTION. (3)

a) $\log_{27} 16 = x \log_3 2$

$$\begin{aligned} \text{LHS} &= \log_{27} 16 \\ &= \frac{\log_3 16}{\log_3 27} \\ &= \frac{\log_3 2^4}{3} \\ &= \frac{4}{3} \log_3 2 \end{aligned}$$

oo $x = \frac{4}{3}$

b) $\frac{\cos 2x}{(\cos x + \sin x)^3} = \frac{\cos x - \sin x}{1 + \sin 2x}$

$$\begin{aligned} \text{LHS} &= \frac{\cos 2x}{(\cos x + \sin x)^3} \\ &= \frac{\cos^2 x - \sin^2 x}{(\cos x + \sin x)^3} \end{aligned}$$

$$= \frac{(\cos x + \sin x)(\cos x - \sin x)}{(\cos x + \sin x)^3}$$

$$= \frac{\cos x - \sin x}{(\cos x + \sin x)^2}$$

$$= \frac{\cos x - \sin x}{\cos^2 x + 2\sin x \cos x + \sin^2 x}$$

b) continued.

$$= \frac{\cos x - \sin x}{1 + \sin 2x}$$

$$= \text{RHS}$$

c) 2 women 1 man = ${}^5C_2 \times {}^4C_1$
+ 2 men 1 woman = ${}^4C_2 \times {}^5C_1$

$$= 10 \times 4 + 6 \times 5$$

$$= \underline{\underline{70 \text{ ways}}}$$

d)

i) $\underline{\underline{1000(1.09)^{35} = 20413.97}}$

ii) 1st yr = $1000(1.09)$

$$\begin{aligned} 2^{\text{nd}} \text{ year} &= 1000(1.09) \\ &\quad + (1000)(1.09)^2 \end{aligned}$$

$$\begin{aligned} 3^{\text{rd}} \text{ yr} &= 1000(1.09) + 1000(1.09)^2 \\ &\quad + 1000(1.09)^3 \end{aligned}$$

$$35^{\text{th}} \text{ yr} = 1000(1.09 + 1.09^2 + \dots + 1.09^{35})$$

$$= 1000 \left(\frac{a(r^n - 1)}{r - 1} \right) \quad \begin{aligned} a &= 1.09 \\ r &= 1.09 \\ n &= 35 \end{aligned}$$

$$= 1000 \left(\frac{1.09(1.09^{35} - 1)}{1.09 - 1} \right)$$

$$= \underline{\underline{\$235,124.72}}$$

(B:2) 2007 Yr II Yearly-continuers: SECTION B.

QUESTION 3 (CONTINUED).

e) $2x^3 + 0x^2 + 6x - 1 = 0$

i) $\alpha^3 * \beta^3 * \gamma^3 + \alpha^3 \beta^2 \gamma^3 + \alpha^2 \beta^3 \gamma^3$
 $= \alpha^2 \beta^2 \gamma^2 (\alpha \beta + \alpha \gamma + \beta \gamma)$
 $= (\alpha \beta \gamma)^2 (\alpha \beta + \alpha \gamma + \beta \gamma)$

$$\alpha \beta \gamma = -\frac{d}{a} = \frac{1}{2}.$$

$$\alpha \beta + \alpha \gamma + \beta \gamma = \frac{c}{a} = \frac{b}{2} = 3.$$

$$\therefore (\alpha \beta \gamma)^2 * (\alpha \beta + \alpha \gamma + \beta \gamma) = \left(\frac{1}{2}\right)^2 * 3$$
$$= \underline{\underline{\frac{3}{4}}}.$$

ii) $\frac{\alpha}{\beta \gamma} + \frac{\beta}{\alpha \gamma} + \frac{\gamma}{\alpha \beta}$

$$= \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha \beta \gamma}$$

$$= \frac{(\alpha + \beta + \gamma)^2 - 2(\alpha \beta + \beta \gamma + \alpha \gamma)}{\alpha \beta \gamma}.$$

$$\Rightarrow \alpha + \beta + \gamma = -\frac{b}{a} = 0.$$

$$\alpha \beta + \beta \gamma + \alpha \gamma = \frac{c}{a} = 3.$$

$$\alpha \beta \gamma = -\frac{d}{a} = \frac{1}{2}.$$

e) ii) CONTINUED

$$= \frac{0 - 3}{\frac{1}{2}}.$$

$$= \underline{\underline{-6}}$$

B:3) 2007 YR11 - Yearly - Continuers - SECTION B.

QUESTION 4.

Let $y = x\sqrt{x^2+3}$.

$$\frac{dy}{dx} = vu' + uv'$$

where $u = x$ $u' = 1$.

$$v = (x^2+3)^{\frac{1}{2}} \quad v' = \frac{1}{2}(2x)(x^2+3)^{-\frac{1}{2}}$$

$$\begin{aligned} \frac{dy}{dx} &= (x^2+3)^{\frac{1}{2}}(1) + (x)(x)(x^2+3)^{-\frac{1}{2}} \\ &= \sqrt{x^2+3} + \frac{x^2}{\sqrt{x^2+3}} \end{aligned}$$

$$= \frac{x^2+3+x^2}{\sqrt{x^2+3}}$$

$$= \frac{2x^2+3}{\sqrt{x^2+3}}$$

$$= \frac{ax^2+b}{\sqrt{x^2+3}}$$

$a=2$ $b=3$.

B) No of ways around a table

$$= (n-1)!$$

$$= 5!$$

$$= 120 \text{ ways}$$

b) (continued)

ii) $\underline{1} \quad \underline{{}^4C_1} \quad \underline{{}^3C_1} \quad \underline{{}^3C_1} \quad \underline{{}^2C_1} \quad \underline{1}$

$\uparrow$
fix.

$$= 1 \times 4 \times 3 \times 3 \times 2 \times 1$$

$$= \underline{\underline{72 \text{ ways}}}$$

c) $\tan 2\theta = \tan \theta$.

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = \tan \theta$$

$$2 \tan \theta = \tan \theta - \tan^3 \theta$$

$$\tan^3 \theta + \tan \theta = 0$$

$$\tan \theta (\tan^2 \theta + 1) = 0$$

either:

$\tan^2 \theta + 1 = 0$ which can't happen so

$$\tan \theta = 0$$

thus this happens when

$$\sin \theta = 0$$

General solution is.

$$\theta = n\pi \quad n \in \mathbb{Z}$$

QUESTION 4 (continued)

d) i) $x = -at$ $A(3at, -at^2)$

$$x^2 = -9ay$$

$$y =$$

gradient of tangent

$$\frac{dy}{dx} = \frac{2x}{-9a} = m,$$

gradient of normal is.

$$m_2 = -\frac{1}{m_1}$$

$$= \frac{9a}{2x} \text{ @ } (3at, -at^2)$$

Point gradient formula.

$$y - y_1 = m(x - x_1)$$

$$y + at^2 = \frac{9a}{2(3at)} (x - 3at)$$

$$baty + 6a^2t^3 = 9ax - 27a^2t$$

$$6ty + 6at^3 = 9x - 27at$$

$$2ty + 2at^3 = 3x - 9at$$

$$\underline{3x - 2ty = 2at^3 + 9at}$$

d) ii)

$$3x - 2ty = 2at^3 + 9at$$

$$\text{at } x = -at$$

$$-3(at) - 2ty = 2at^3 + 9at$$

$$-2ty = 2at^3 + 9at + 3at$$

$$y = \frac{2at^3 + 12at}{-2t}$$

$$y = -at^2 - 6a$$

Co-ordinates of B

$$\underline{\underline{(-at, -at^2 - 6a)}}$$

4e) i) $AC \Rightarrow$

$$\tan 40 = \frac{2h}{AC}$$

$$AC = \frac{2h}{\tan 40}$$

$$\underline{\underline{\tan 40}}$$

$AD \Rightarrow$

$$\tan 50 = \frac{3h}{AD}$$

$$AD = \frac{3h}{\tan 50}$$

$$\underline{\underline{\tan 50}}$$

3:5) 2007 Yr11 Yearly - Continuers - SECTION B.

QUESTION 4 (CONTINUED)

e) continued.

ii). By Pythagoras.

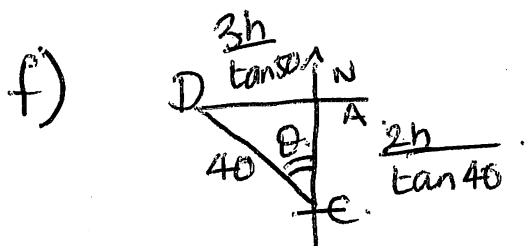
$$40^2 = \left(\frac{2h}{\tan 40} \right)^2 + \left(\frac{3h}{\tan 50} \right)^2$$

$$40^2 = \frac{4h^2}{\tan^2 40} + \frac{9h^2}{\tan^2 50}$$

$$40^2 = h^2 (4 \cot^2 40 + 9 \cot^2 50)$$

$$h^2 = \frac{40^2}{4 \cot^2 40 + 9 \cot^2 50}$$

$$h = \frac{40}{\sqrt{4 \cot^2 40 + 9 \cot^2 50}}$$



$$\begin{aligned} \tan \theta &= \frac{3h}{\tan 50} \div \frac{2h}{\tan 40} \\ &= \frac{3h}{\tan 50} \times \frac{\tan 40}{2h} \\ &= \frac{3}{2} \frac{\tan 40}{\tan 50} \end{aligned}$$

$$\tan \theta = 1.0561$$

$$\theta = 46.56$$

∴ Bearing of D from C
is $360^\circ - 46.56^\circ$

$$= 313^\circ \text{ (nearest degree)}$$